Project Report: Library Management System

1. Project Overview

The **Library Management System** is a console-based Python application designed to automate the basic operations of a local library. It allows administrators to maintain a digital record of inventory, track book availability, and manage the lifecycle of a book from addition to borrowing and eventual return.

2. Objectives

- To create a centralized digital repository for book titles.
- To implement real-time status tracking (Available vs. Borrowed).
- To provide a user-friendly command-line interface (CLI) for library operations.

3. System Features

The system is built around a Library class that encapsulates the following functionalities:

- **Inventory Management:** Add new titles to the collection or remove existing ones (provided they aren't currently borrowed).
- **Status Tracking:** Automatically assigns and updates the status of books using a Python dictionary for $O(1)$ lookup efficiency.
- **Search & Visibility:** Users can search for specific titles or view the entire catalog, including a total count of books.
- **Circulation Logic:** Handles the logic for borrowing and returning books, preventing users from borrowing books that are already out.